AI/ML RESEARCH METHODOLOGY BLUEPRINT

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Project Title / Topic: Neighbourhood-aware misbehavior detection for V2X networks (VeReMi Extension benchmark)

1. PROBLEM FORMULATION:

- Specific ML/AI Problem or Failure Mode Addressed:

Connected vehicles broadcast position, speed and heading. A misbehaving vehicle sends false data (random or spoofed position, frozen speed) that misleads others. Most learned detectors judge each sender **in isolation**, so evidence from the surrounding vehicles goes unused, and it is untested how a neighbour-aware detector behaves when many neighbours are themselves malicious (cooperative schemes often assume an honest majority). Failure mode: missed or slow detection of falsified data, and unknown robustness to malicious neighbours.

2. HYPOTHESIS FORMULATION:

- Null Hypothesis (H0):

A detector that also attends to the 16 nearest neighbours has the same F1 as the per-sender baseline on held-out traffic (F1 full = F1 per-sender).

- Alternative Hypothesis (H1):

The neighbourhood-aware detector has higher F1 (F1 full > F1 per-sender), with higher recall and shorter detection latency and no increase in false positives. Secondary (H1b): its recall stays within about 2 points of its clean value even when over half of a sender's neighbours are malicious (H0b: recall falls as malicious neighbours increase).

3. RESEARCH & EXPERIMENTAL DESIGN:

- Dataset to be used (e.g., ImageNet, Kaggle, Custom collected):

VeReMi Extension (public benchmark, Zenodo; simulated traffic and attackers, F2MD/VEINS/SUMO, Luxembourg scenario, 30% attackers, 19 attack types). Done: RandomPos and ConstSpeed (14:00-16:00). Planned: other attacks, dense 07:00-09:00 subsets, MixAll_0024, VeReMi NextGen. Windows: 16 nearest senders x 10 s (61,708 train / 11,251 val / 26,255 test).

- Baseline Model vs. Proposed Architecture/Modification:

Baseline: per-sender transformer (no neighbour information); also a no-spatial ablation. **Proposed:** factorised temporal + spatial transformer in which each sender attends over its own history and over the other senders in the window.

- Evaluation Metrics (e.g., F1-Score, mAP, BLEU, AUC-ROC):

- **Precision:** of the senders we flag as malicious, the share that really are (few false alarms).
- **Recall:** of all malicious senders, the share we catch (few missed attackers).
- **F1-score:** balance of precision and recall (their harmonic mean); one-number summary of detection quality.
- **False-positive rate (FPR):** share of honest senders wrongly flagged as malicious.
- **AUC-ROC:** how well the model ranks malicious senders above honest ones over all thresholds (1.0 perfect, 0.5 random).
- **Detection latency:** seconds from a sender's first appearance to the first alarm raised on it.
- **Recall/FPR by malicious-neighbour share:** checks the detector still works when many of a sender's neighbours are attackers.

4. PROPOSED VALIDATION STRATEGY:

- (e.g., K-Fold Cross Validation, Statistical Significance Test):

- **Done:** train on the first hour, test on the later hour; receiver-disjoint validation/test blocks; threshold chosen on validation only; 3 seeds per model (mean and std). No formal significance test yet.
- **Planned:** grouped 5-fold cross-validation by receiver vehicle (5-10 seeds); paired bootstrap 95% CIs on the F1 difference; Wilcoxon signed-rank test across folds/seeds and McNemar's test on paired decisions; Holm-Bonferroni correction across attack types; colluding-attacker stress test; low-to-high density transfer.

5. EXPECTED AI / COMPUTATIONAL OUTCOME:

- The neighbour-aware model is expected to beat the per-sender baseline by about 1 F1 point, mainly through recall. Preliminary (2 attacks, 3 seeds): F1 0.982 vs 0.973, recall 0.975 vs 0.959, precision 0.989 vs 0.987, FPR about 0.5% for both.
- Faster alarms (preliminary mean latency 0.15 s vs 0.70 s) and recall near 0.97 even with over 50% malicious neighbours (0.971 vs 0.973 clean).
- Some attacks may not benefit: ConstSpeed was slightly worse with neighbours (0.948 vs 0.960), so H0 should be rejected overall, not necessarily for every attack.
- Small model (about 0.6 MB, about 10 min CPU training). Scope: simulated benchmark, independent attackers; no collusion or crash intent.